

B. Tech Degree V Semester Examination, November 2008

ME 502 METROLOGY AND MACHINE TOOLS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A (Answer all questions)

(8 x 5 = 40)

- I. (a) Distinguish between Accuracy and Precision.
(b) Describe various gauge materials and wear allowance.
(c) Write a short note on clinometer.
(d) What is optical flat? Explain its uses.
(e) How an engine lathe is specified? Explain with a block diagram.
(f) Explain the working of a Radial Drilling Machine
(g) Differentiate between Honing and Lapping
(h) Explain salient features of an NC machine.

PART – B

(4 x 15 = 60)

- II. (a) Explain various sources of errors in measurement system. (10)
(b) Explain line and end measurements. (5)
- OR**
- III. (a) Explain various types of fits used for the purpose of assembly of machine parts. (10)
(b) Write a short note on tolerance and allowance. (5)
- IV. (a) Describe the working of an auto collimator with the help of a neat sketch. (10)
(b) Explain the uses of sine bar. (5)
- OR**
- V. (a) Describe the RMS and CLA methods of estimating roughness value. (10)
(b) Describe the symbolic representation of surface roughness as per BIS. (5)
- VI. (a) Describe any five work holding devices used in lathe. (10)
(b) Differentiate between Capstan and Turrent lathe. (5)
- OR**
- VII. (a) Explain any one quick return mechanism used in shaping machine. (10)
(b) Write a note on counter boring and counter sinking. (5)
- VIII. (a) Describe various types of milling machines used in manufacturing industries. (10)
(b) Explain various gear finishing operation. (5)
- OR**
- IX. (a) Explain three types of motion control of tools in CNC machines. (10)
(b) Explain nomenclature of swiss type automatic screw machines. (5)

